

Self-Learning: Address Decoding Techniques & 8259 Cascading

MPCA Module 4

> **Definition:** Address Decoding Techniques define how system address lines are mapped to peripheral chip selects (Absolute vs Partial), while **8259 Cascading** expands interrupt handling capacity from 8 to 64 priority channels.

1. Absolute vs Partial Address Decoding

ABSOLUTE ADDRESS DECODING:

- On a system with 16 address lines (A15-A0), the first 4 lines (A15-A12) are used to select one of 16 possible chip selects. This is done by connecting A15-A12 to the CS* pin of the peripheral chip. This method ensures that only one chip is active at any given time.

PARTIAL ADDRESS DECODING:

- On a system with 16 address lines (A15-A0), the first 4 lines (A15-A12) are used to select one of 16 possible chip selects. This is done by connecting A15-A12 to the CS* pin of the peripheral chip. This method allows for multiple chips to be active at the same time, but it requires additional logic to ensure that the correct chip is selected.

Comparison Table:

Feature	Absolute Decoding	Partial Decoding
Number of active chips	1	Multiple
Requires additional logic	No	Yes

2. 8259 Cascaded Mode (Master-Slave Configuration)

To handle more than 8 interrupts, one **Master 8259** is connected to up to **8 Slave 8259s** via the 3-bit Cascade Bus (**CAS_0, CAS_1, CAS_2**).

MASTER 8259: The Master 8259 is connected to the system bus and has its own 8 interrupt lines (IR0-IR7). It also has a 3-bit Cascade Bus (CAS_0, CAS_1, CAS_2) to connect to Slave 8259s.

Cascading Operation Flow:

- Slave 8259 receives interrupt on its IR line and signals Master 8259 via Master's IR pin.
- Master 8259 signals the system bus via its own IR pin.

3. Quick Recall Flow

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